

Problem & Data

The AI Backbone — 10 Essential Topics

From vague goal to trained model

1. Problem Understanding — Watch the Human First

Don't start with "which model?" Start with "what would a HUMAN do?"

Hospital

What decisions
do doctors
struggle with?

Predict disease
Spot abnormal reports
Smart ICU alerts

Online Store

What costs you
money or time
right now?

Predict demand
Detect fraud
Group customers

ExamGuard

What does
cheating LOOK
like?

Phone on desk
Looking sideways
Unusual behavior

Cricket Team

What does the
coach actually
DO?

Analyze opponent
Predict form
Field placement

Technique: Keep asking "but what EXACTLY?" until you get something a human can physically DO.

2. The 3 Questions — The Core Skill

Q1: What DATA?

Eyes = IMAGES
Words = TEXT
Measurements = NUMBERS

Q2: HOW thinking?

Recognizing = SUPERVISED
Something off = ANOMALY
Grouping = CLUSTERING
Balancing = RL

Q3: Past examples?

YES labeled = SUPERVISED
NO just normal = UNSUPERVISED
NO learns trying = RL

IMAGES + Recognize + YES

CNN / YOLO

IMAGES + Abnormal + NO

Autoencoder

TEXT + Categorize + YES

Naive Bayes

NUMBERS + Predict word + YES

Classification

NUMBERS + Predict number + YES

Regression

NUMBERS + Abnormal + NO

Isolation Forest

NUMBERS + Group + NO

K-Means / DBSCAN

ANY + Balance + Rewards

RL

Practice: Restaurant Manager — Apply the 3 Questions

Task	Q1: Data	Q2: Thinking	Q3: Labels?	ML Type
Predict chicken to buy	NUMBERS (past sales)	Predicting amount	YES past sales	Regression
Categorize reviews	TEXT (reviews)	Good/bad/ average	YES past reviews	Naive Bayes
Unusual electricity bill	NUMBERS (bills)	Doesn't match normal	NO just knows normal	Anomaly Detection
Set lunch prices daily	NUMBERS+ JUDGMENT	Balance profit vs customers	NO learns by results	RL
Spot theft on CCTV	IMAGES (camera)	Recognize cash in hand	YES knows theft	CNN/YOLO

3. Data Collection — 4 Sources (In Order)

1. Already Have

Check databases,
CCTV, logs, records

Hospital records
Website analytics
Past exam footage

ALWAYS check first!

2. Public Dataset

Kaggle, Roboflow,
HuggingFace

50K+ free datasets
Labeled images
Text corpora

Search before collecting!

3. Pre-trained

YOLO, MediaPipe,
BERT, Whisper

Zero data needed!
Download and test
in 10 minutes

Fastest path to v1!

4. Collect Yourself

Camera, forms,
scraping, sensors

Last resort
Match real setup
Min 200 samples

Only if 1-3 fail!

4. Data Validation — 5 Checks Before Using Any Dataset

1. SIZE

Count rows/images
Enough for model?

<500 risky
500-5K good
10K+ great

2. LOOK

Open 30 random samples
Clear? Labels correct?

Blurry = remove
Wrong label = fix

3. RELEVANCE

Matches YOUR situation?
Street phone != desk phone

Camera angle must
match deployment

4. BALANCE

Equal per class?
500+500 = good
4500+50 = bad

Imbalanced =
model ignores minority

5. TRUST

Downloads? Comments?
Ratings? Others used it?

1000+ downloads
= probably decent

5. Data Augmentation — 500 Images → 3000+

Flip

Mirror
horizontally

Phone left
vs right

Rotate

Tilt
5-30 degrees

Camera
tilted

Brightness

Lighter
or darker

Morning vs
evening

Zoom

Closer
or further

Different
distances

Noise

Add grain
or blur

Cheap camera
simulation

Crop

Different
framing

Phone partially
visible

1 image x 6 techniques = 6 images. 500 original = 3000 augmented. But REAL data > augmented data.

6. Data Cleaning — 80% of ML Engineer's Time

Missing Values

Delete row OR
fill with average

Duplicates

Find and
remove

Wrong Labels

Check 50 random
fix mistakes

Outliers (age=500)

Detect, remove
or cap

Wrong Format

'Rs 50,000' →
50000

Inconsistent

'Male','M','m'
→ standardize

Blurry Images

Remove or
recapture

Wrong File Type

.bmp → .jpg
convert

7. Pre-trained Models — Zero Data to Start

YOLOv8

Detect 80 objects
(phone, person, car...)

Data: ZERO

pip install
ultralytics

MediaPipe Face

Face landmarks
gaze direction

Data: ZERO

pip install
mediapipe

MediaPipe Pose

Body skeleton
33 keypoints

Data: ZERO

Same
package

BERT

Text understanding
sentiment, classify

Data: ZERO

pip install
transformers

Whisper

Speech to text

Data: ZERO

pip install
openai-whisper

The 10-Photo Test (10 minutes):

Take 10 photos from YOUR setup → run pre-trained → check results:

>85% correct = USE AS-IS | 60-85% = FINE-TUNE with 300 photos | <60% = wrong model, try another

8. Data Labeling — How to Label Properly

Classification Label

Whole image =
'phone' or 'no phone'

Is there a phone?
Simplest type

Bounding Box

Draw rectangle
around the phone

WHERE is the phone?
YOLO needs this

Segmentation

Outline exact shape
pixel by pixel

Medical imaging
Self-driving (precise)

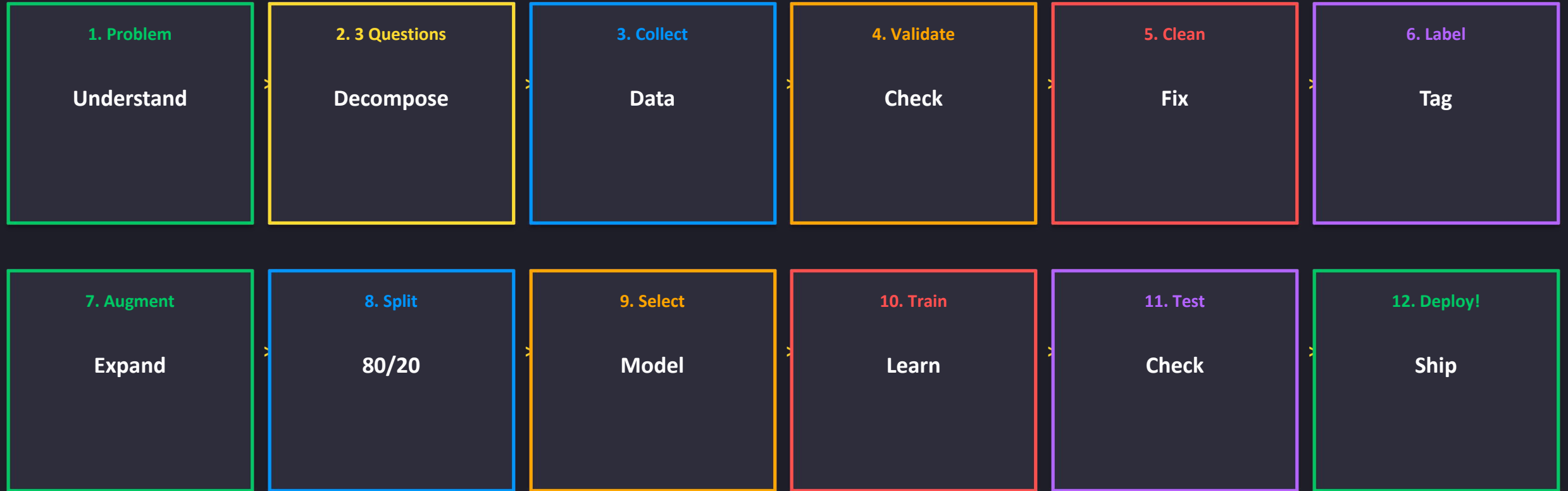
Text Label

'positive' / 'negative'
per review

Review sentiment
Ticket routing

Tools: Roboflow (web) | Labellmg (desktop) | CVAT (video) | Label Studio (all types)

9. Data Pipeline — Complete A-to-Z Flow



Time: 60% data work | 20% training | 10% testing | 10% deployment. DATA IS THE JOB.

10. Multi-Model Systems — Real AI Architecture

Real systems = multiple independent models + regular code connecting them

